

Existence and Regularity of Competitive Equilibria in Economies with Externalities

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Overview

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1. Introduction & Motivation

- **Economic Interconnectedness:** Externalities represent the core of real-world interactions where an individual's choices directly affect others' preferences and expenditures.
- **Welfare & Market Failures:** External effects often result in inefficient resource allocation, exacerbating wealth disparities, educational inequalities, and environmental challenges (e.g., global industrial pollution as noted by Sandmo, 2011).
- **Other-Regarding Preferences:** Households facing negative externalities may distortively alter behaviors (e.g., higher short-run consumption, reduced savings, longer work hours; Dupor & Liu, 2003).
- **Objective of the Paper:** Rigorously examine a pure exchange economy under broad forms of externalities using **differential topology techniques** rather than traditional fixed-point arguments.

2. Research Gap & Contributions

The Definition of a Regular Economy

An economy is deemed **regular** if it possesses a *finite (odd) number of equilibria*, and each equilibrium exhibits local, differentiable dependence on the underlying parameters.

- **The Research Gap:** Traditional literature (Bonnisseau 2003; Bonnisseau & Del Mercato 2010; Balasko 2015) relies heavily on restrictive geometric preference properties or ignores consumption constraints under externalities.
- **Our Contribution:**
 - Establishes generic regularity under consumption and wealth concerns in a differential setup.
 - Utilizes an elegant **homotopy approach** to demonstrate existence.
 - Employs the **Transversality Theorem** to prove that regular economies form an open and full-measure subset.

Framework: Goods, Prices, and Parameters

- **Commodity Space:** There are L finite distinguishable goods, represented by $\mathbb{R}^L$.
- For any bundle $x = (x^1, \dots, x^L) \in \mathbb{R}^L$, let $x^\backslash = (x^1, \dots, x^{L-1}) \in \mathbb{R}^{L-1}$ be the non-numéraire coordinates.
- **Price System:** Strictly positive prices due to scarcity. Choosing commodity L as the *numéraire* ($p_L = 1$), the price space is:

$$\mathbb{S} = \mathbb{R}_{++}^{L-1} \times \{1\}$$

- **Consumers & Endowments:** There are m agents. Each consumer i has an endowment $e_i \in X = \mathbb{R}^L$.
- Total parameter space is $\Omega = X^m$. The vector $b = (p, e_1, \dots, e_{m-1}, w_m) \in \mathbb{B}$ uniquely describes prices, endowments of the first $m - 1$ agents, and the wealth of the last agent.

Utility Functions and Standard Assumptions

Each agent i possesses a price-dependent and endowment-dependent utility function $u_i : \mathbb{R}_{++}^{Lm} \times \mathbb{S} \times \Omega \rightarrow \mathbb{R}$.

Assumption 1 (Smooth Classical Properties)

For all $i = 1, \dots, m$:

- ① u_i is $\mathcal{C}^3$ on $\mathbb{R}_{++}^{Ll} \times \mathbb{S} \times \Omega$.
- ② For each $(x_{-i}, p, e) \in \mathbb{R}_{++}^{L(l-1)} \times \mathbb{S} \times \Omega$, the function $u_i(\cdot, x_{-i}, p, e)$ is differentiable strictly increasing, i.e., for every $x_i \in \mathbb{R}_{++}^L$, $D_{x_i} u_i(x_i, x_{-i}, p, e) \in \mathbb{R}_{++}^L$.
- ③ For each $(x_{-i}, p, e) \in \mathbb{R}_{++}^{L(l-1)} \times \mathbb{S} \times \Omega$, the function $u_i(\cdot, x_{-i}, p, e)$ is differentiable strictly quasi-concave, i.e., for every $x_i \in \mathbb{R}_{++}^L$, $D_{x_i}^2 u_i(x_i, x_{-i}, p, e)$ is negative definite on $\text{Ker } D_{x_i} u_i(x_i, x_{-i}, p, e)$.

Utility Functions and Standard Assumptions

Assumption 2 (Boundary Behavior Extension)

For all $i = 1, 2, \dots, I$, there exists a differentiable extension of $\tilde{u}_i$ on $\mathbb{R}_{++}^L \times \mathbb{R}_+^{L(I-1)} \times S \times \Omega$ such that for each $(x_i, x_{-i}, q, e) \in \mathbb{R}_{++}^L \times \mathbb{R}_+^{L(I-1)} \times S \times \Omega$,

- (i) $cl_{\mathbb{R}^L} \{x'_i \in \mathbb{R}_{++}^L : \tilde{u}_i(x'_i, x_{-i}, q, r) \geq \tilde{u}_i(x_i, x_{-i}, q, r)\} \subseteq \mathbb{R}_{++}^L$;
- (ii) $D_{x_i} \tilde{u}_i(x, q, r) \in \mathbb{R}_{++}^L$.

Points (ii) simply means that the extension of the utility function is strictly increasing, and point (i) implies that if the consumption of some commodities becomes very small, then the amount of some other commodities have to become very large in order to remain the same utility level.

Utility Functions and Standard Assumptions

To guarantee the generic regularity of economies with externalities, we impose a structural macro-behavior constraint on the matrices of derivatives.

Assumption 3

Let $(x, e, v, p) \in \mathbb{R}_{++}^{Lm} \times \Omega \times \mathbb{R}^{Lm} \times \mathbb{S}$. If for all consumer i , $v_i \in \text{Ker } D_{x_i} u_i(x_i, x_{-i}, p, e)$ and the gradient $D_{x_i} u_i(x_i, x_{-i}, p, e)$ is proportional to the prices vector p , then the following property is satisfied for all consumers $k = 1, \dots, m$.

$$v_k D_{e_k x_k}^2 u_k(x_k, x_{-k}, p, e)(v_k) + v_k D_{x_k}^2 u_k(x_k, x_{-k}, p, e)(v_k) < 0 \text{ whenever } v_k \neq 0, \\ (v_i D_{x_k x_i}^2 u_i(x_i, x_{-i}, p, e) + v_i D_{e_k x_i}^2 u_i(x_i, x_{-i}, p, e))(v_k) = 0 \text{ for all } i \neq k.$$

- **Intuition:** The second-order welfare impact of own consumption structurally dominates the aggregated external effects.
- Automatically satisfied in traditional economies without externalities or models with additively separable utility functions (e.g., Geanakoplos & Polemarchakis, 2008).

Competitive Equilibrium

Define the space of endogenous variables as $\Xi := \mathbb{R}_{++}^{Lm} \times \mathbb{R}_{++}^m \times \mathbb{R}_{++}^{L-1}$, containing elements $\xi := (x, \lambda, p^\backslash)$.

Definition 1 (Equilibrium System)

A bundle (x^*, p^*) along with $\lambda^* \in \mathbb{R}_{++}^m$ is a competitive equilibrium if $F_e(\xi^*) = 0$, where $F_e : \Xi \rightarrow \mathbb{R}^{\dim \Xi}$ is defined by:

$$F_e(\xi) = \left(\left(F_e^{(i,1)}(\xi), F_e^{(i,2)}(\xi) \right)_{i=1,\dots,m}, F_e^M(\xi) \right) = 0$$

where:

$$F_e^{(i,1)}(\xi) := D_{x_i} u_i(x_i, x_{-i}, p, e) - \lambda_i p$$

$$F_e^{(i,2)}(\xi) := -p \cdot (x_i - e_i)$$

$$F_e^M(\xi) := \sum_{i=1}^m x_i^\backslash - \sum_{i=1}^m e_i^\backslash$$

Definition 2(Regular economies)

$e \in \Omega$ is a regular economy if 0 is a regular value of the mapping F_e , that is for each $\xi^* \in F_e^{-1}(0)$, the differential mapping $D_\xi F_e(\xi^*)$ is onto.

Existence Theorem (Theorem 1)

Theorem

A competitive equilibrium exists for any economy $(u, e) \in \mathcal{U} \times \Omega$.

Proof Strategy via Homotopy Methods:

- 1 Construct a simple reference system $L(\xi) = 0$ matching an isolated standard classical economy (no externalities) at fixed q^* and x^{**} , which has a unique known solution.
- 2 Define the continuous deformation homotopy $H : \Xi \times [0, 1] \rightarrow \mathbb{R}^{\dim \Xi}$:

$$H^{(i,1)}(\xi, t) = D_{x_i} u_i(x_i, tx_{-i} + (1-t)e_{-i}, p_t, e) - \lambda_i p$$

$$H^{(i,2)}(\xi, t) = -p \cdot (x_i - (te_i + (1-t)x_i^{**}))$$

- 3 Clearly, $H(\xi, 0) = L(\xi)$ and $H(\xi, 1) = F_e(\xi)$.
- 4 Prove that the zero-set $H^{-1}(0)$ is sequentially compact using boundary behavior (Assumption 2). By Homotopy Invariance, $F_e^{-1}(0) \neq \emptyset$.

Regularity Results (Theorem 2)

Define the extended global mapping $F : \Xi \times \Omega \rightarrow \mathbb{R}^{\dim \Xi}$ where $F(\xi, e) := F_e(\xi)$.

Lemma

0 is a regular value for the extended global map $F(\xi, e)$.

Proof Sketch.

Show that the total Jacobian matrix $D_{\xi, e}F(\xi^*, e^*)$ has full row rank. Suppose $\Delta \cdot D_{\xi, e}F = 0$. Expanding this system, orthogonality implies $\Delta x_i \in \ker D_{x_i} u_i$. Applying Assumption 3, we obtain a structural contradiction unless $\Delta x_k = 0, \forall k$. This immediately forces $\Delta \lambda = 0$ and $\Delta p^\backslash = 0$, yielding $\Delta = 0$. $\square$

Theorem

The set of extended regular economies $\mathcal{R}^$ is an **open and full Lebesgue measure subset** of Ω .*

Key Geometric Properties

From the Regular Value Theorem and the Transversality Theorem, we instantly obtain the structure of the equilibrium manifold:

Corollary 1 (Local Structure of Regular Economies)

For any given regular economy $(u, e) \in \mathcal{U} \times \mathcal{R}_u$:

- ❶ **Finiteness:** $F_e^{-1}(0) = \{\xi_1, \dots, \xi_r\}$ is finite (and odd).
- ❷ **Stability:** There exists an open neighborhood $V(e) \subset \Omega$, and local C^1 maps $g_k : V \rightarrow U_k(\xi_k)$ such that for any perturbed economy $e' \in V$:

$$F_{e'}^{-1}(0) = \{g_1(e'), \dots, g_r(e')\}$$

- ❸ **Differentiable dependence:** The equilibria move smoothly with respect to perturbations in parameters, allowing robust comparative statics.

Conclusion & Policy Implications

- **Theoretical Core:** Validated both existence and generic stability of competitive equilibria under comprehensive, non-separable consumption and wealth externalities.
- **Mathematical Rigor:** Bypassed standard fixed-point constraints through a well-behaved differential homotopy approach.
- **Economic Insight:** Established that regular economies are generic (dense and full measure), confirming that chaotic or unstable equilibria are rare anomalies.
- **Policy Implications:** Provides a robust foundation for classic comparative statics. Governments can predictably deploy instruments like **commodity taxes**, **equal transfers**, or emission quotas to achieve Pareto improvements without risking localized structural collapse.